Calling Agent / Auto-Dialer

Full feature specification & build guide for CreatorAI Studio

"Like Tata dialer" — cloud dialer + autonomous AI voice agent · 6 September 2026

How to read this. Part 1 is the complete feature catalogue of a calling-agent product in the class of Tata Smartflo / Exotel / Ozonetel — every capability, grouped. Each feature is tagged BUY (use a vendor) or BUILD (we implement it). Part 2 is the architecture and data model to build it inside this codebase. Part 3 is the phased delivery plan and cost model. Part 4 is the India compliance checklist. Part 5 is risks and the decisions you need to make.

0 · Two layers, one product

A modern "calling agent" is two stacked capabilities. Build both; ship the AI layer first because it is the one our self-serve SMB users can actually use without hiring a call-centre team.

Layer	What it is	Who talks on the call	Primary users
1 · Auto-dialer / cloud contact centre	Software places calls from a list and manages inbound calls, queues, agents, IVR, reporting — the "Tata dialer" core.	A human agent	Businesses with a sales/support team
2 · Autonomous AI voice agent	An AI (speech-to-text → LLM → text-to-speech) holds the conversation, qualifies, books, confirms, escalates.	The AI (can warm-transfer to a human)	Everyone — "press a button, the AI calls your 300 new leads"

PART 1 — FULL FEATURE CATALOGUE

A · Telephony foundation

Feature	Detail	Tag
Phone number provisioning	Buy/port DIDs; India **140-series** (promotional) and **1600-series** (BFSI/transactional) numbers; toll-free; virtual numbers per campaign/brand.	BUY
Caller-ID management	Set outbound CLI per campaign; caller-ID rotation / pooling to spread volume; branded caller ID (CNAM / Truecaller business verification).	BUILD logic on BUY numbers
Carrier / SIP connectivity	PSTN termination + origination, SIP trunk, WebRTC gateway. Concurrency (channels) limits.	BUY
Call recording	Full-call or on-demand; dual-channel (separate agent/customer tracks) for QA/transcription; pause/resume for PCI/PII moments; encrypted storage; retention policy; legal-hold.	BUY capture · BUILD policy/retention UI
Recording storage & lifecycle	Store to our object storage (Supabase/S3), signed URLs, auto-delete after N days, per-org retention config.	BUILD
Voicemail / answering-machine handling	Answering-Machine Detection (AMD / beep detection); "voicemail drop" of a pre-recorded message; skip-to-next on machine.	BUY AMD · BUILD drop flow
Local presence dialing	Automatically dial from a number in the lead's own area/region to lift pickup rate.	BUILD on BUY number pool

B · Outbound dialer engine (the core of "Tata dialer")

Campaign management

- **Campaign object** — name, owner, contact list / lead segment, script or AI assistant, caller ID(s), schedule (date range + daily window), timezone handling, pacing mode, concurrency, retry policy, disposition set, priority.
- **List / lead management** — import CSV / connect a saved lead segment; de-dupe; field mapping; per-record custom fields available to the script; list priority ordering; recycle rules (re-add "no answer" after N hours).
- **Contact scrubbing** — remove DND/DNC registrations, internal suppression list, already-contacted-today, invalid numbers, and `do_not_contact` leads *before* the list is dialable.
- **Scheduling** — start/stop times, per-day windows, blackout dates (holidays), timezone-aware so nobody is called outside 09:00–21:00 local.
- **Retry / attempt rules** — max attempts per lead; wait between attempts; different retry gaps per outcome (busy → 15 min, no-answer → 3 h, voicemail → next day); "give up" disposition after last attempt.
- **Caller-ID rotation** — cycle through a number pool; cap calls/number/day to avoid spam-flagging.
- **Campaign controls** — start / pause / resume / stop / archive; live progress (dialed / connected / remaining / % complete); clone a campaign.

Dialer modes (pacing)

Mode	Behaviour	Needs
Preview	Agent sees the lead record first, clicks to dial. Full control, lowest throughput.	1 agent : 1 call
Progressive	Auto-dials the next number the instant an agent is free. No abandoned calls.	1 : 1
Power / Ratio	Dials a fixed multiple of free agents (e.g. 2× or 3×). Some risk of abandoned calls.	agent pool
Predictive	Uses pickup-rate + talk-time statistics to dial *ahead* of agent availability; maximises talk time; must keep abandon rate under a legal threshold.	larger agent pool
Parallel / "AI blast"	Fire many simultaneous calls; the ones that answer are handed to an **AI agent** (unlimited "agents"). This is the mode that fits our product.	AI voice agent + provider concurrency

Tag: BUILD the pacing/orchestration engine; it drives a BUY telephony API.

On-connect handling

- Route answered call → free human agent (with screen-pop) OR → AI voice agent OR → play an IVR / message.
- Whisper to the agent before connect ("Lead: Priya, source: Diwali landing page, score 82").
- "No agent available" fallback — queue the customer, offer callback, or drop with an apology + SMS.

Disposition & wrap-up

- Mandatory **disposition code** at end of every call (Interested / Not interested / Callback / Wrong number / DNC / No answer / Voicemail / Failed / Converted). Configurable per campaign.
- **After-call work (ACW) / wrap-up timer** — agent time to add notes before next call; max wrap time.
- Auto-actions per disposition (create task, send WhatsApp, move lead stage, add to suppression).
- **Scheduled callbacks** — "call me at 4 pm tomorrow"; personal callback (same agent) vs any-agent; callback reminders.

C · Inbound call handling

- **Multi-level IVR** — DTMF + spoken-input menus, nested; business-hours / after-hours / holiday branches; language selection.
- **ACD (Automatic Call Distribution)** — queue inbound calls, distribute to agents by rule.
- **Routing strategies** — round-robin, longest-idle, simultaneous ring, skills-based, priority, **sticky agent** (same agent the lead spoke to last), VIP routing.
- **Queue experience** — position announcement, estimated wait, hold music / marketing messages, **callback-in-queue** ("keep my place, call me back"), max-queue-time overflow to voicemail.
- **Voicemail** — per-agent / per-queue mailboxes, voicemail-to-email, voicemail transcription.
- **Missed-call service** — auto SMS/WhatsApp on missed call; auto-create a lead.
- **Inbound → same pipeline** — an inbound caller who isn't a known lead becomes a lead (source = inbound call).

Tag: BUY IVR/ACD primitives from the CPaaS · BUILD the visual flow builder + routing config UI.

D · Agent workspace

- **WebRTC softphone in the browser** — no desk phone; also mobile app / SIP option.
- **Screen-pop / CTI** — the lead record opens automatically on connect, with history, notes, AI briefing.
- **Call controls** — answer, hang up, hold, mute, **blind & warm transfer**, **3-way conference**, add participant, DTMF send, record pause/resume.
- **Call scripting** — campaign script shown in-call, with branching, dynamic fields (`{name}`, `{offer}`), objection-handling snippets, required-disclosure blocks.
- **Agent states / AUX codes** — Ready, On call, Wrap-up, Break (with reason: lunch / training / bio), Offline; state history for reporting; auto-not-ready after missed call.
- **Notes & disposition panel**, tags, next-action, follow-up scheduling.
- **Personal callback list**, do-not-call toggle, "request supervisor".

Tag: BUILD the workspace UI · BUY the WebRTC/SIP client SDK.

E · Supervisor & Quality Assurance

- **Live wallboard** — calls in progress, in queue, longest wait, abandon rate, agents by state, campaign progress, SLA %, service level.
- **Real-time agent board** — who's on a call / wrapping / on break, call duration, current campaign.
- **Monitor · Whisper · Barge · Takeover** — listen silently; coach the agent unheard by the customer; join as a 3-way; take the call over.
- **Manual QA** — scorecards, evaluation forms, calibration, agent acknowledgement, dispute flow.
- **Automated QA (AI)** — score *100%* of calls against a rubric: script adherence, mandatory disclosures spoken, tone/empathy, dead-air, talk-over, compliance phrases, resolution. Flag outliers for human review.
- **Coaching** — annotate recordings, assign coaching sessions, track improvement.

Tag: BUILD dashboards + QA UI · BUY monitor/whisper/barge primitives + speech analytics (or run our own via Claude on transcripts).

F · Click-to-call, number masking, call patching

- **Click-to-call** from the lead record / any list — "connect two numbers": ring the agent, then ring the lead, bridge them.
- **Number masking / privacy** — agent and lead never see each other's real number; a proxy number bridges them (marketplace-style privacy, useful for field sales / gig).
- **Call patching** with pre-call whisper of lead context to the agent.
- **Mobile-agent mode** — dial the agent's own mobile, then patch — works with no softphone / poor internet (common in India field sales).

Tag: BUY "connect two numbers" API · BUILD the buttons + masking number lifecycle.

G · CRM & integrations

- **Native** — calls, recordings, transcripts, dispositions, callbacks write straight onto our own `leads` / notes / tasks.
- **Screen-pop** into the CreatorAI Studio lead page.
- **Outbound webhooks** — `call.started/ended`, `recording.ready`, `transcript.ready`, `disposition.set` → customer's own systems.

- **Inbound API** — start a campaign, add numbers to a running campaign, click-to-call, fetch CDRs/recordings.
- **"API dialplan"** equivalent — on an inbound call, hit the customer's URL to decide routing from their data.
- **Zapier / Make / n8n** connectors; two-way sync to external CRMs (Zoho, HubSpot, LeadSquared, Salesforce) — later.

Tag: BUILD (fits the existing automations + social-integration patterns).

H · Reporting & analytics

- **CDR / call log** — every call: direction, numbers, agent, campaign, times, duration, wait, hold, disposition, recording, cost, hang-up cause.
- **Campaign reports** — dialed, contacted, connect rate, ASR (answer-seizure ratio), abandon rate, avg talk time, disposition breakdown, conversions, cost per contact / per conversion, list penetration.
- **Agent performance** — calls handled, AHT (avg handle time), talk vs wrap vs idle, occupancy, adherence, first-call resolution, QA score, conversions, revenue.
- **Queue / SLA** — service level (% answered in X s), avg speed of answer, abandonment, callback take-up.
- **Dashboards** — live + historical; filter by date / campaign / agent / disposition; drill to the call + recording.
- **Scheduled exports** — email / WhatsApp a daily/weekly PDF or CSV; API pull; data warehouse sync.

Tag: BUILD (extends the existing Analytics module).

I · Conversation intelligence (AI on every call)

- **Transcription** — full transcript, speaker-separated, timestamped; multilingual + Hinglish; searchable.
- **AI call summary** — reason for call, what was said, commitments made, sentiment shift, next step — written to the lead note (Claude, already in our stack).
- **Auto-disposition** — AI proposes the disposition code from the transcript; agent confirms.
- **Sentiment & emotion** — per call and trend over time; real-time "customer getting frustrated" alert to a supervisor.
- **Topic / keyword spotting** — competitor mentions, price objections, churn signals, compliance keywords, profanity.
- **Auto QA scoring** — every call scored (section E).
- **Next-best-action** — "send the pricing PDF on WhatsApp", "book a demo", "mark lost".
- **Coaching insights** — phrases top performers use, talk-ratio, filler words.

Tag: BUILD using Claude + a transcription API (Deepgram / Sarvam / provider-native).

J · Autonomous AI voice agent (the modern differentiator)

- **Real-time voice loop** — streaming STT → LLM → streaming TTS, < 800 ms round-trip; **barge-in** (agent stops instantly when the human speaks); voice-activity detection; back-channelling ("mm-hm").
- **Multilingual, India-first** — Hindi, Tamil, Telugu, Bengali, Marathi, Kannada, Gujarati, Punjabi, Hinglish; per-lead language auto-select; code-switching.
- **Script = brand-aware prompt** — objective + opening line + talking points + FAQs + objection handling + guardrails, drafted by Claude from the brand profile (reuse `avatarScript.ts` pattern), editable by the user.
- **Fixed compliance preamble** — first utterance is a non-editable disclosure: "This is an automated AI assistant from <brand>. This call may be recorded." (India-mandated.)
- **In-call tools / function calling** — `mark_interested`, `mark_not_interested`, `schedule_callback(datetime)`, `mark_do_not_contact`, `send_whatsapp(template)`, `check_calendar_slot`, `answer_from_knowledge_base(q)`, `transfer_to_human(reason)`.
- **Knowledge base / RAG** — upload FAQs / catalogue / pricing; the agent answers from real docs, cites nothing it wasn't given (same discipline as the existing chatbot).
- **Warm transfer / human handoff** — on request, complexity, or anger: bridge to an available human with a spoken context summary; or, if none available, book a callback task and notify the assignee.
- **Voicemail drop** — detect machine, leave a natural pre-generated message, schedule a retry.
- **Guardrails** — no promises on price/discount beyond configured limits; profanity / abuse handling; "I'm not able to help with that" fallback; max call duration; loop detection.
- **Post-call** — transcript + recording + AI summary + proposed disposition + sentiment written back to the lead; automation trigger fires.
- **Testing** — "call me now to test" mode; simulated-call sandbox; A/B test two scripts on a split of the list; per-variant connect/interest rates.
- **Voice selection** — pick the agent voice/persona per brand; consistent across calls.

Tag: **BUY** the voice-agent platform (Bolna / Vapi / Retell) · **BUILD** script authoring, tools, campaign glue, results.

K · Compliance & governance (India — mandatory)

- **DLT registration** — Principal Entity + telemarketer + headers + **call scripts/templates** registered on the TRAI DLT platform before any dial.
- **Number-series enforcement** — promotional calls only from 140-series; BFSI/transactional from 1600-series.
- **Calling-hours guard** — 09:00–21:00 in the lead's timezone; hard block outside.
- **DND / DNC scrubbing** — real-time check against the TRAI registry + per-org suppression list; auto-add on "do not call".
- **Consent capture & audit** — documented, timestamped opt-in stored at lead creation (web form / landing page / WhatsApp); calls blocked without a valid consent reference; consent ledger exportable for audit.
- **Mandatory AI disclosure** + **recording disclosure** as the fixed opening line.
- **Frequency capping** — max calls per lead per day / week.
- **Data residency** — India-resident storage option for recordings/transcripts; DPDP-compliant processing; PII redaction in stored transcripts; configurable retention + right-to-erasure.
- **Audit log** — who launched which campaign to which list, when, with what script.

- **Abandon-rate cap** for predictive mode; auto-throttle if breached.

Tag: BUILD every gate · BUY DND-registry access + (optionally) a CPaaS that fronts DLT for customers.

L · Platform & admin

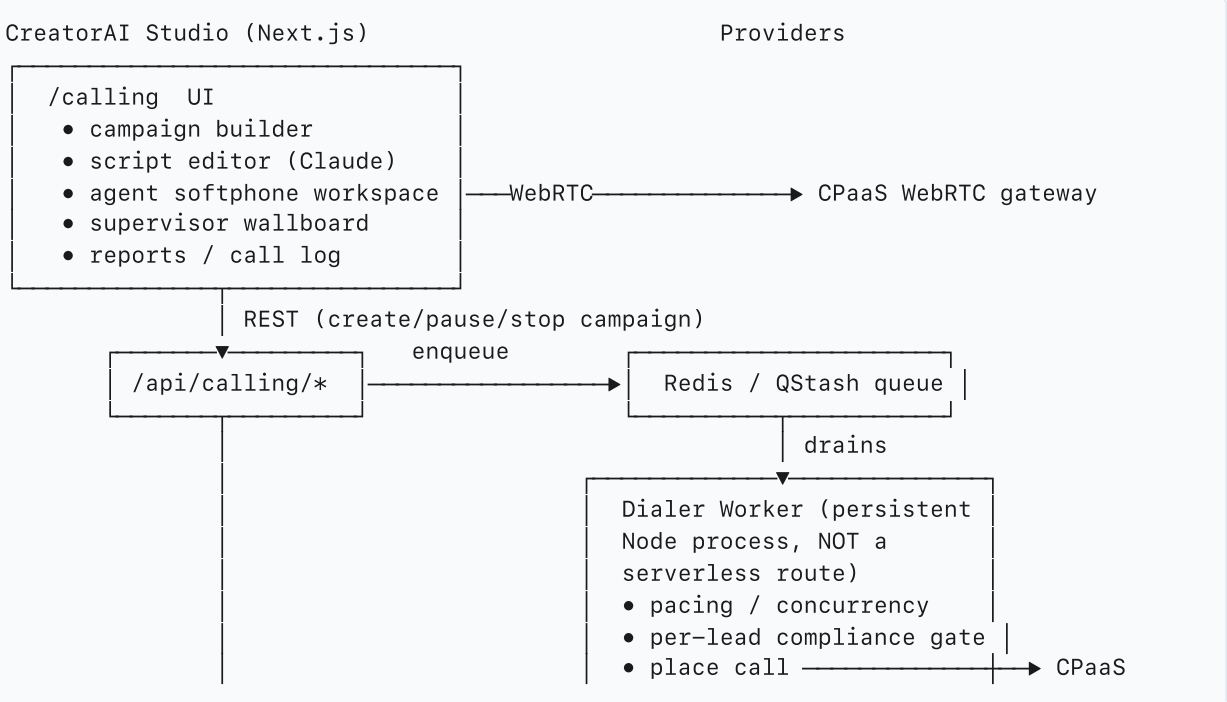
- Multi-tenant (already have orgs); agent/user management & licences; number inventory per org.
- **Per-minute metering & billing** — calls are billed by the minute, not per unit → needs its own meter separate from the current per-generation credits; usage caps; auto-recharge; cost alerts.
- Rate limits & concurrency caps per org; fair-use.
- Roles — who can launch campaigns, who can only monitor, who can spend call credits.
- Status page / call-quality metrics (MOS, jitter, packet loss) per call.

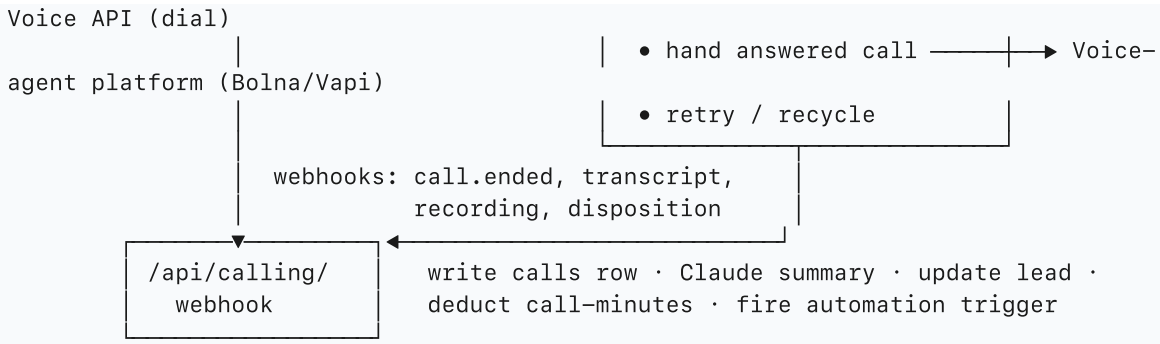
PART 2 — HOW TO BUILD IT INTO CREATORAI STUDIO

2.1 Build vs buy — per layer

Layer	Decision	Why
PSTN / SIP / numbers / recording / AMD / WebRTC	**BUY** a CPaaS	Regulated, carrier-grade, never build. India: **Exotel** or **Ozonetel** (they also help with DLT); **Plivo** for a clean API; Twilio if multi-country.
Autonomous voice agent (STT+LLM+TTS realtime)	**BUY** a voice-agent platform	**Bolna** (Indian-language first, bulk campaigns, HITL, India data residency) as primary; **Vapi** as alt (keeps Claude as the LLM, BYO TTS/telephony).
Transcription / speech analytics	**BUY** API	Deepgram / Sarvam (Indic) / provider-native. Feed transcripts to Claude for summary, QA, sentiment.
Campaign engine, pacing, retry, scheduling, scrubbing	**BUILD**	This is the product. A queue + worker orchestrating the CPaaS.
Compliance gates (consent, DND, hours, disclosure, caps)	**BUILD**	Legal exposure is ours; must be enforced in our code, not hoped for.
UI — campaigns, agent workspace, supervisor wallboard, reports	**BUILD**	Next.js, fits existing dashboard.
Script authoring, in-call tools, results write-back, automation triggers	**BUILD**	Reuses Claude + leads + automations already in the codebase.
Metering / call-credits billing	**BUILD**	Per-minute meter alongside existing credits.

2.2 Architecture





> **The load-bearing constraint.** A calling campaign is a long-running, stateful, rate-limited job. A Next.js serverless route cannot host the dialer loop. You need **a persistent worker + a queue** — e.g. a small Node service (BullMQ + Redis / Upstash) on a always-on host (Railway, Fly, a VM), or a cron that drains a `calls` table where `status='queued'`. *Decide this once for both the dialer and the Module 23 video/ffmpeg jobs.*

2.3 Data model (new migration)

```
call_providers        org_id, kind (cpaas|voice_agent|transcription), name,
                      credentials_ref, is_active, config jsonb

call_numbers          org_id, provider, e164, series (140|1600|tollfree|virtual),
                      label, dlt_registered bool, caps jsonb (per-day limit),
                      is_active

call_assistants       org_id, name, language[], voice_id, objective,
                      opening_preamble (locked), script jsonb (talking points,
                      FAQs, objections, guardrails), tools jsonb,
                      knowledge_base_ref, created_by

call_campaigns        org_id, name, assistant_id, mode (preview|progressive|
                      power|predictive|parallel), lead_filter jsonb (segment),
                      caller_number_ids[], schedule jsonb (window, tz, blackout),
                      retry_policy jsonb, concurrency, dispositions jsonb,
                      status (draft|running|paused|done), stats jsonb,
                      created_by

call_queue_items      campaign_id, lead_id, attempt, next_attempt_at,
                      state (queued|dialing|done|skipped), skip_reason

calls                 org_id, campaign_id, lead_id, assistant_id, direction,
                      provider, provider_call_id, from_e164, to_e164,
                      status (queued|ringing|in_progress|completed|failed|
                      no_answer|busy|voicemail|abandoned),
                      started_at, answered_at, ended_at, duration_sec,
                      wait_sec, hold_sec, hangup_cause,
                      recording_url, transcript jsonb, ai_summary,
                      sentiment, disposition, disposition_by (ai|agent),
                      qa_score jsonb, agent_id, cost_minutes, cost_amount,
                      consent_ref, compliance jsonb (preamble_spoken, dnd_checked)

call_callbacks        org_id, lead_id, campaign_id, scheduled_at, agent_id
                      (nullable = any), status, note

call_suppression      org_id, e164, reason (dnd|dnc|opt_out|invalid|manual),
```

```
                      source, created_at

call_agents           org_id, user_id, extension, skills[], status
                      (ready|on_call|wrap|break|offline), status_reason,
                      status_since

agent_status_events   agent_id, status, reason, at            (for adherence)

call_qa_scorecards    org_id, name, rubric jsonb
call_qa_reviews       call_id, scorecard_id, scores jsonb, reviewer (ai|user_id),
                      total, flagged bool, notes

call_consents         org_id, lead_id, channel, captured_at, evidence_ref,
                      scope (calls|recording|marketing), revoked_at
```

Extend existing: `leads` already has `do_not_contact`, `preferred_channel`, `phone`, notes, tasks — reuse them. Add `call_minutes_balance` (or a `call_credits` ledger) to the org/profile.

2.4 API routes to add

```
POST /api/calling/campaigns              create / update
POST /api/calling/campaigns/:id/start    validate + enqueue
POST /api/calling/campaigns/:id/pause
POST /api/calling/campaigns/:id/stop
GET  /api/calling/campaigns/:id/stats    live progress
POST /api/calling/assistants             CRUD; ?draft=1 → Claude drafts script from
brand
POST /api/calling/click-to-call          { leadId }  → connect agent ⇔ lead
POST /api/calling/callbacks              schedule / list
GET  /api/calling/calls                  CDR list + filters
GET  /api/calling/calls/:id              detail + transcript + recording
POST /api/calling/webhook/:provider      ← provider events (signed)
POST /api/calling/agent/status           agent sets Ready/Break/etc.
GET  /api/calling/live                   supervisor wallboard feed (SSE)
POST /api/calling/monitor                { callId, mode: listen|whisper|barge }
POST /api/calling/suppression            add / import DNC
```

2.5 Provider abstraction (mirror `src/lib/social/*`)

```
src/lib/calling/
  provider.ts        // interface: placeCall, transferCall, hangup,
                     //            getRecording, monitorCall, listNumbers
  bolna.ts           // primary voice-agent provider
  vapi.ts            // alt voice-agent provider
  exotel.ts          // CPaaS: numbers, click-to-call, AMD, recordings
  compliance.ts      // consentOk(lead), dndScrub(list), withinHours(lead),
                     //   frequencyOk(lead), buildPreamble(brand)
  pacing.ts          // preview|progressive|power|predictive|parallel logic
  disposition.ts     // codes + per-disposition auto-actions
  script.ts          // Claude: brand profile -> editable spoken script
```

Every provider file exports `isXConfigured()`; the `/calling` UI shows a disabled state with a one-line reason when unconfigured — identical to the pattern already used for HeyGen, Replicate, Meta, etc.

2.6 What it reuses from the current codebase

- **Leads pipeline** — segments feed campaigns; outcomes update `leads` + notes + tasks.
- **Claude** (`src/lib/ai/*`) — script drafting, call summary, auto-disposition, sentiment, QA scoring.
- **Automations engine** — add trigger types `call_completed`, `call_disposition_changed`, `callback_due`; existing actions (task, notify, WhatsApp, SMS, email) work as-is.
- **Brand profile / banned words** — feed the call script exactly like content/chatbot.
- **Notifications** — "AI needs a human on line 2", "callback due", "campaign finished".
- **Credits** — add a per-minute meter next to the existing per-generation credits.
- **Provider-gating + disabled-UI pattern**, OAuth-state helper, storage buckets for recordings.
- **Analytics module** — call reports slot in beside the existing metrics.

PART 3 — DELIVERY PLAN

3.1 Phases & feature mapping

Phase	Scope	Features from Part 1
0 · Spike ~1 wk	One CPaaS + one voice-agent (Bolna trial). Hardcoded assistant. Call one test number end-to-end; webhook updates a lead. Validate Hindi / Hinglish quality on real recordings.	A (minimal), J (minimal), K (preamble only)
1 · MVP ~4–6 wk	`call_campaigns` + `calls` + queue + **persistent worker**. Segment → campaign → parallel/progressive dial → AI agent runs a Claude-drafted editable script → webhook writes disposition + transcript + summary → lead updated → call-minutes deducted. Compliance gates: consent check, DND scrub, 9–21 window, frequency cap, fixed AI+recording preamble. Basic results dashboard + call log + recordings.	B (campaigns, list mgmt, scrubbing, scheduling, retry, parallel/progressive, disposition, callbacks), I (transcript, summary, auto-disposition), J (voice loop, multilingual, script, preamble, basic tools, post-call), K (all gates), H (call log + campaign report), L (per-minute metering)
2 · Depth ~4–6 wk	In-call tools (callback booking, WhatsApp follow-up, KB answers, transfer-to-human). Voicemail drop. Retry/recycle rules per outcome. Automation triggers. Knowledge-base RAG. A/B script testing. Sentiment + AI auto-QA on every call. Scheduled report exports.	J (tools, RAG, transfer, voicemail drop, A/B, guardrails), I (sentiment, auto-QA, next-best-action), B (advanced retry), H (scheduled exports), G (webhooks out)
3 · Human agents ~6–8 wk	WebRTC softphone workspace, screen-pop, call controls, scripts, agent states, wrap-up. Supervisor wallboard + monitor/whisper/barge. Preview/power/predictive modes. Click-to-call + number masking. Skills routing. Manual QA scorecards.	D (all), E (all), B (preview/power/predictive), F (all), C (basic ACD/queue)
4 · Contact-centre grade ongoing	Full multi-level IVR builder, callback-in-queue, sticky agent, external-CRM two-way sync, Zapier, data-warehouse export, MOS/quality metrics, India data-residency option, advanced adherence reporting, DLT-registration onboarding wizard.	C (full), G (external CRM), H (warehouse), K (residency), L (quality)

3.2 Ownership

Sits naturally with **Stream 4 (External Integrations)** for the telephony + voice-agent provider wiring, with **Stream 1** owning the schema + metering and **Stream 2** owning the Claude script/summary/QA prompts. The persistent-worker infrastructure is a shared decision with the Module 23 video jobs — assign one person to own "background job platform" across both.

3.3 Rough effort

Assumes buying CPaaS + voice-agent platform, not building realtime audio.

- Phase 0: 1 engineer × 1 week
- Phase 1 (MVP): 2 engineers × 4–6 weeks

- Phase 2: 2 engineers × 4–6 weeks
- Phase 3 (human agents): 2–3 engineers × 6–8 weeks
- Phase 4: ongoing

MVP that a customer can actually run a campaign on: **~6–8 weeks** from a standing start.

3.4 Cost model (India, domestic mobile, outbound)

Component	Rate
Telephony (local CPaaS)	₹0.35–0.75 / connected min (Twilio ≈ ₹0.65)
AI voice agent (STT+LLM+TTS)	₹1–3 / connected min (Bolna ≈ ₹4–5/1,000 min platform + model cost)
Transcription (if separate)	₹0.2–0.5 / min
Blended, connected minutes	**≈ ₹1.5–4 / min**
Ring-no-answer	telephony pennies only — AI meter doesn't start
One-time integration (self-serve API, us)	engineering time only; not the ₹1–5 lakh "managed project" figure

Worked example — 1,000 leads, ~35% connect, ~2 min avg AI talk

- 350 connected calls × 2 min = 700 billed minutes
- 700 × ₹3 blended ≈ **₹2,100** provider cost for the campaign
- Price to the customer at, say, ₹8–12/min or a "calling credits" pack → healthy margin

Metering: bill the customer per *connected minute* (round up to 30 s or 1 min), show a live spend counter, hard-stop at their cap, offer auto-recharge. Keep this separate from the existing per-image/video credits.

PART 4 — INDIA COMPLIANCE CHECKLIST (do not skip)

1. □ Business (Principal Entity) + telemarketer entity registered on TRAI **DLT**.
2. □ Every call **script/template** registered & approved on DLT before use.
3. □ Outbound CLI is a **140-series** number (promotional) / **1600-series** (transactional/BFSI).
4. □ Dialer refuses any number outside **09:00–21:00** in the lead's timezone.
5. □ Real-time **DND/DNC** scrub against the TRAI registry + org suppression list on every lead before dial.
6. □ **Documented, timestamped consent** stored per lead (form / landing page / WhatsApp opt-in); dial blocked without a valid `consent_ref`.
7. □ AI agent's **first utterance** = fixed "automated AI + call may be recorded" disclosure (not model-generated, not skippable).
8. □ **Frequency cap** enforced (max calls / lead / day & week).
9. □ Recording storage: encrypted, retention policy, right-to-erasure, optional India residency; **PII redaction** in stored transcripts.
10. □ **Audit log**: who ran which campaign, to which list, with which script, when.
11. □ Predictive mode (if built) keeps **abandon rate** under threshold with auto-throttle.
12. □ Product copy says "call your *consenting* leads" — never "upload any list and cold-call".
13. □ Onboarding step to help the customer get DLT-registered / a 140-series number (or use a CPaaS that fronts this).

Penalties for getting this wrong: up to ₹10 lakh per violation, number blocking, telemarketer-ID blacklisting, DLT suspension.

PART 5 — RISKS & DECISIONS

Risks

- **Background-job platform** — no serverless route can run the dialer loop; needs a persistent worker + queue. Shared with Module 23.
- **Regulatory** — DLT + consent + AI disclosure are hard gates; a shortcut here is an existential risk, not a bug.
- **Indian-language voice quality** — the make-or-break UX factor; validate with real recordings in Phase 0 before committing to a provider.
- **Cost runaway** — a mis-configured campaign can burn money fast; hard caps + live spend counter + auto-pause are MVP, not phase 2.
- **Deliverability / spam-flagging** — number rotation, caps per number, branded caller ID matter for connect rate.
- **Provider lock-in** — the `src/lib/calling/provider.ts` abstraction keeps Bolna/Vapi/Exotel swappable.

Decisions needed

1. Ship order — AI voice agent first (recommended), or human dialer first?
2. Voice-agent provider — **Bolna** (Indic-first, fastest) vs **Vapi** (keeps Claude, more control)?
3. CPaaS — Exotel / Ozonetel (front DLT for customers) vs Plivo (cleaner API, customer handles DLT)?

4. Do we get our own 140-series number pool + DLT, or require the customer to?
5. Metering — per-minute "calling credits" vs a paid add-on plan tier?
6. Background-worker host — Railway / Fly / VM / Upstash QStash? (decide with Module 23)
7. Phase-1 use cases — just "qualify new leads", or also reminders / confirmations / re-engagement / payment nudges?

Sources

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